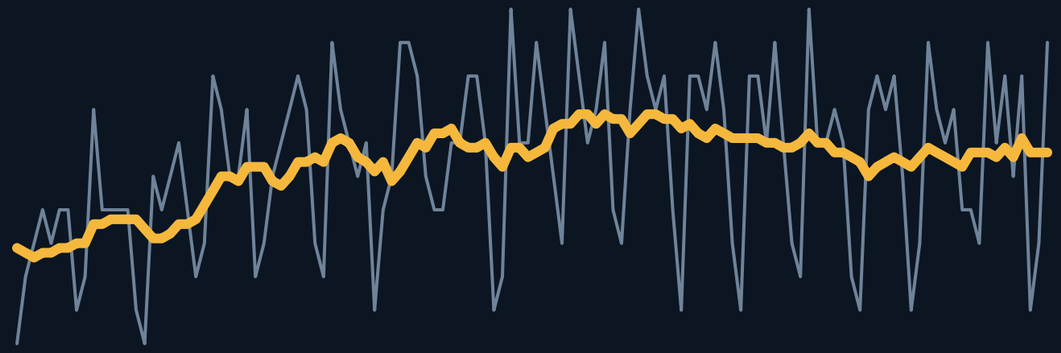
WEEK 1 · DAY 3

Calm Down the Noise

Rolling windows: how to see the real story inside a jumpy, wiggly series. 〰 → ▬



NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

Prepared for NIGCOMSAT

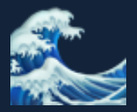
TODAY'S MISSION · 2 HOURS

Three moves



Spot the rhythm

tickets drop every weekend



Smooth it

a 7-day rolling window



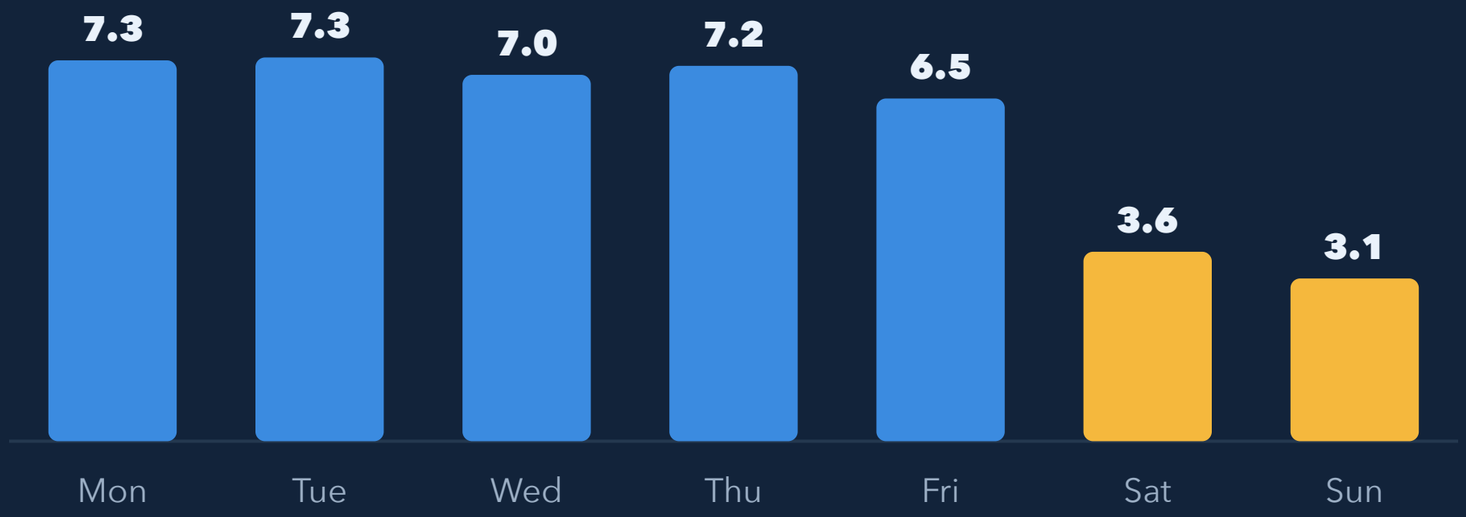
Know what it hides

lag, the future, and short outages

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 60 min hands-on · 🏁 20 min wrap-up

QUICK QUESTION

Which day do faults take a rest? 😴



Average NOC tickets per day, 730 days of data

7.1
tickets on a weekday

3.4
tickets on a weekend day

Even faults rest on Sunday.
Weekends ≈ **half** of weekdays.
😄

PART 1

The moving average 🌊

Average the last few days, then slide forward one day. Repeat.

YOU ALREADY DO THIS

“How much do I spend on fuel?” ⛽

😱

Look at one day

Monday you filled the tank: ₦40,000. Tuesday: ₦0. Panic, then relief. Every day tells a different story.



Average the last 7 days

One big fill-up gets shared across the week. The number stops jumping.

➡️

Slide it every day

Tomorrow, drop the oldest day and add the newest. That's a **rolling (moving) average**.

Same money. **Calmer story.**

HOW IT WORKS

A 7-day window sliding over real tickets

The first 14 days of `noc_tickets_daily`

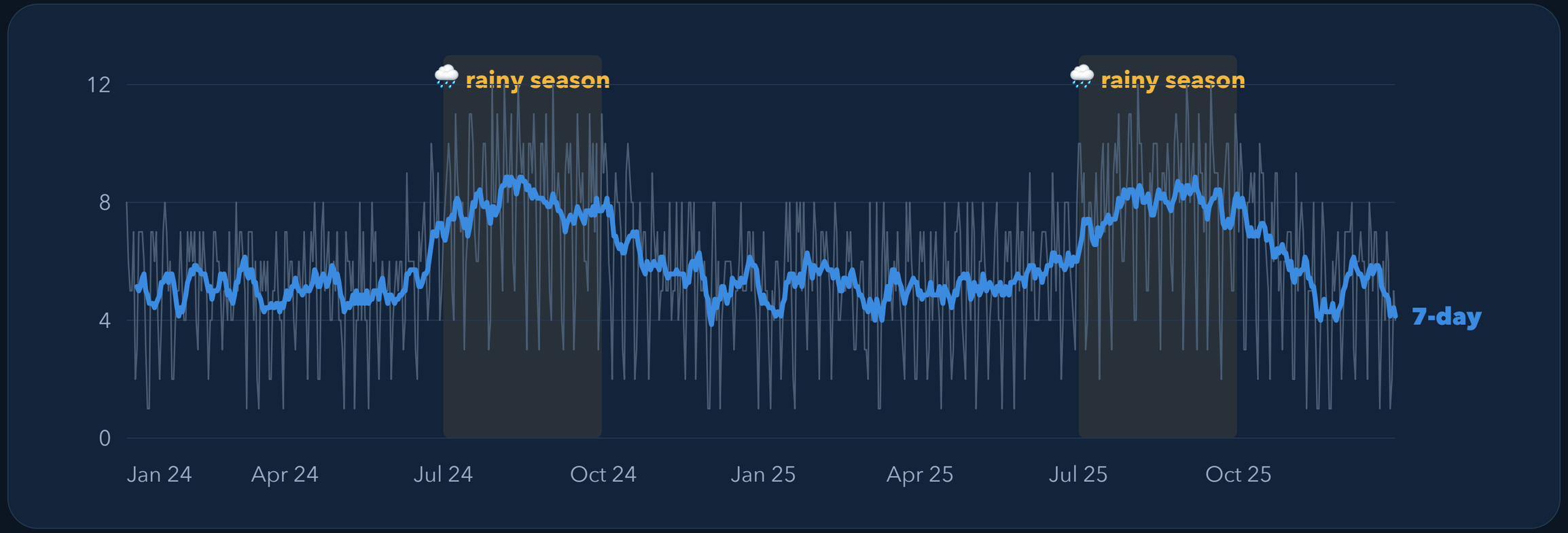
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
8	6	5	5	7	2	3	7	7	7	6	3	1	1
NaN	NaN	NaN	NaN	NaN	NaN	5.1	5.0	5.1	5.4	5.6	5.0	4.9	4.6

```
tk["roll7"] = tk["tickets_raised"].rolling(7).mean()    # gold = 7-day average
```

NaN = not 7 days yet. A 7-day average genuinely doesn't exist until day 7.

THE WHOLE TWO YEARS

Grey = every day. Blue = the 7-day average.



grey zig-zag = weekdays vs weekends

blue = the real level

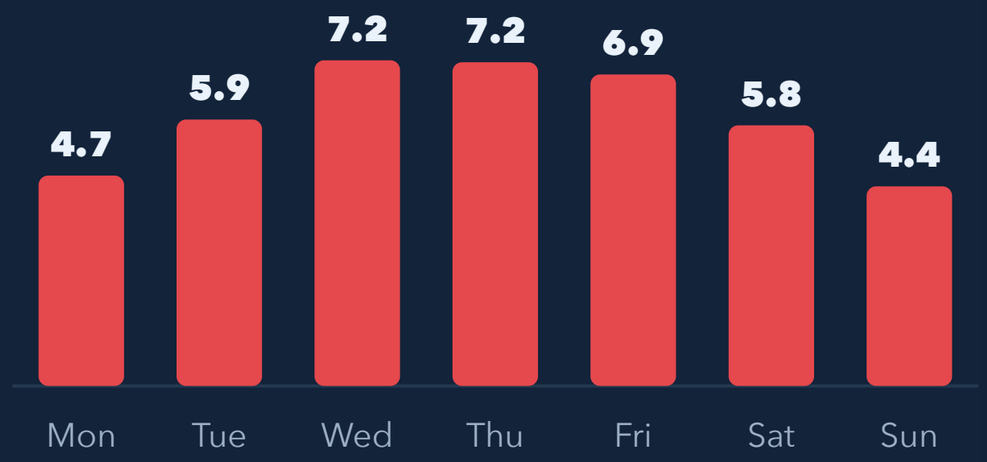
busier every rainy season

WHY 7, AND NOT 3?

7 days = every weekday exactly once

3-day window

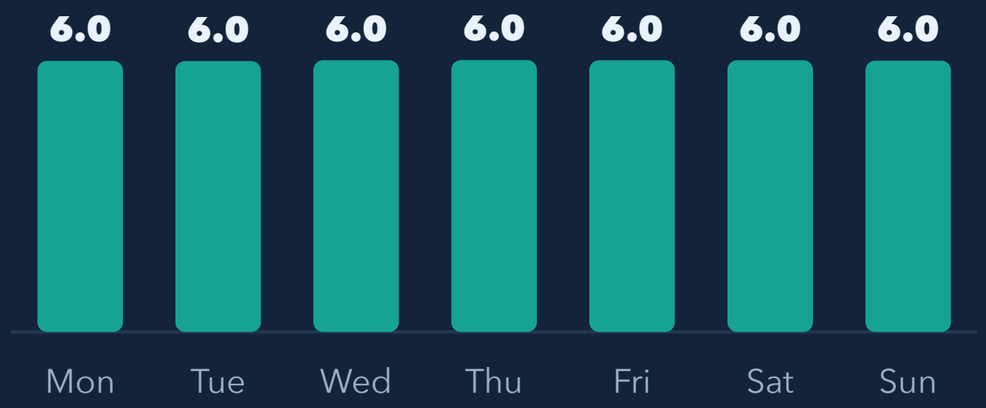
average of the window, by day of week



Still jumps by **2.8** tickets across the week

7-day window

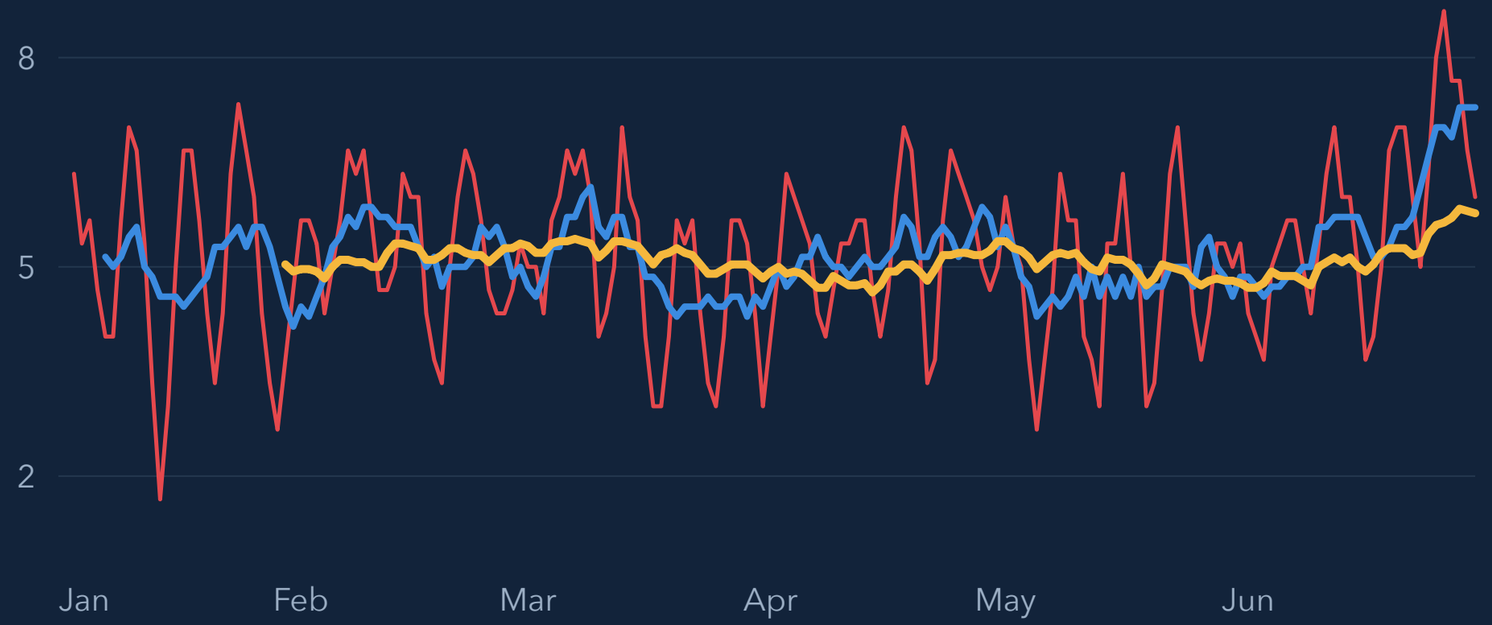
average of the window, by day of week



Jumps by only **0.02**: the weekend effect cancels

PICK YOUR WINDOW

Short = jumpy. Long = calm but slow.



Tickets, Jan to Jun 2024

3 days

still feels the weekend

7 days

removes the weekly rhythm: start here

30 days

shows the season, reacts slowly

PART 2

What smoothing costs 🙈

Every smooth line pays for its calm with something. Know the price.

WHERE DOES THE WINDOW SIT?

Trailing looks back. Centred peeks at the future. 🔮

Trailing window (pandas default)



Uses only the past 30 days. Safe for live monitoring.

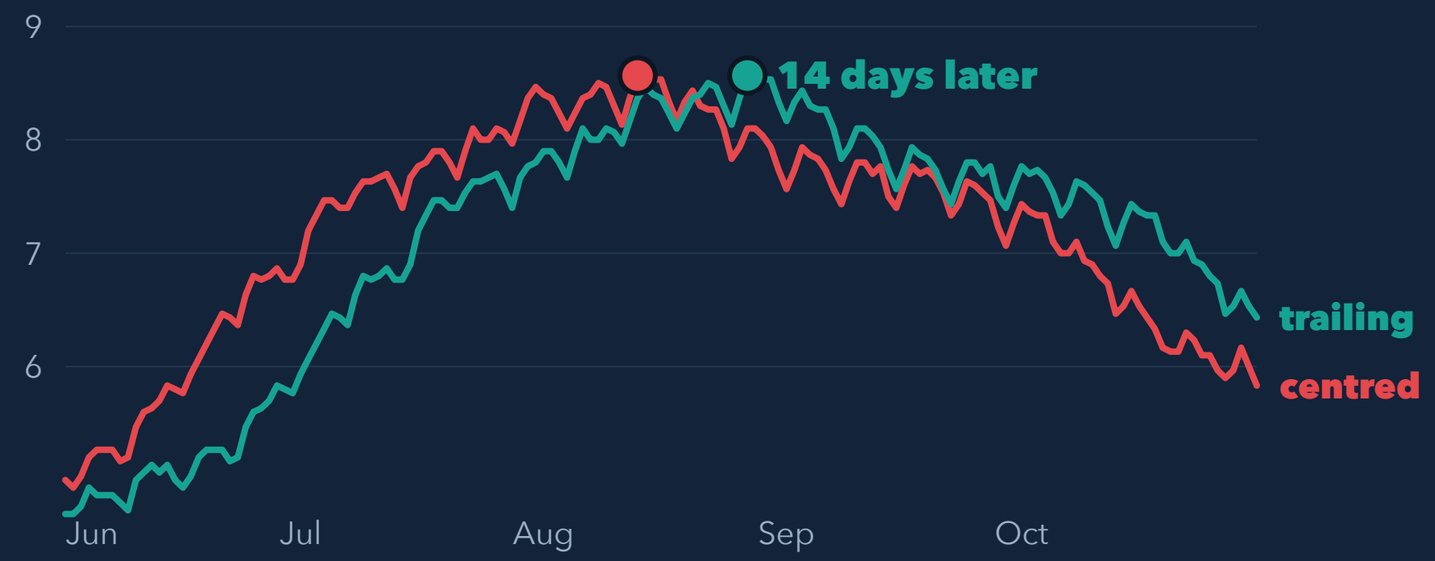
Centred window (`center=True`)



Uses 15 days of the FUTURE. Fine for a report about last year. **Never for live monitoring or forecasts.**

THE PRICE OF THE TRAILING WINDOW

Safe, but about half a window late ⏳



30-day average of tickets, rainy season 2024

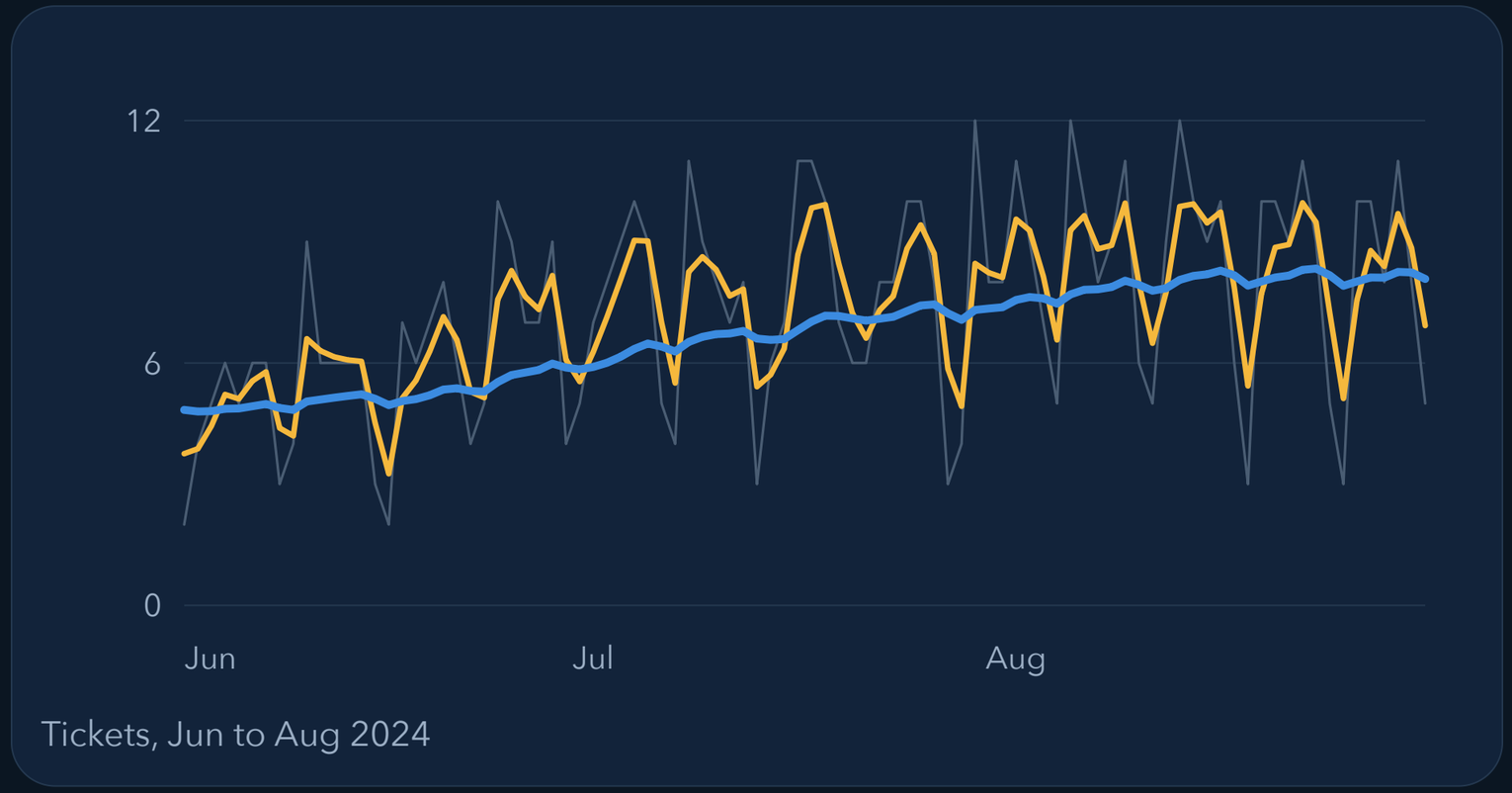
14 days

the trailing peak arrives after the centred one (2024 and 2025 alike)

30-day window → about **15 days** late. Bigger window, bigger delay.

ANOTHER WAY: EXPONENTIAL SMOOTHING

.ewm(alpha) = how fast you believe news



Tickets, Jun to Aug 2024

High alpha (0.5)

believes today's gist immediately: follows the data closely, reacts fast

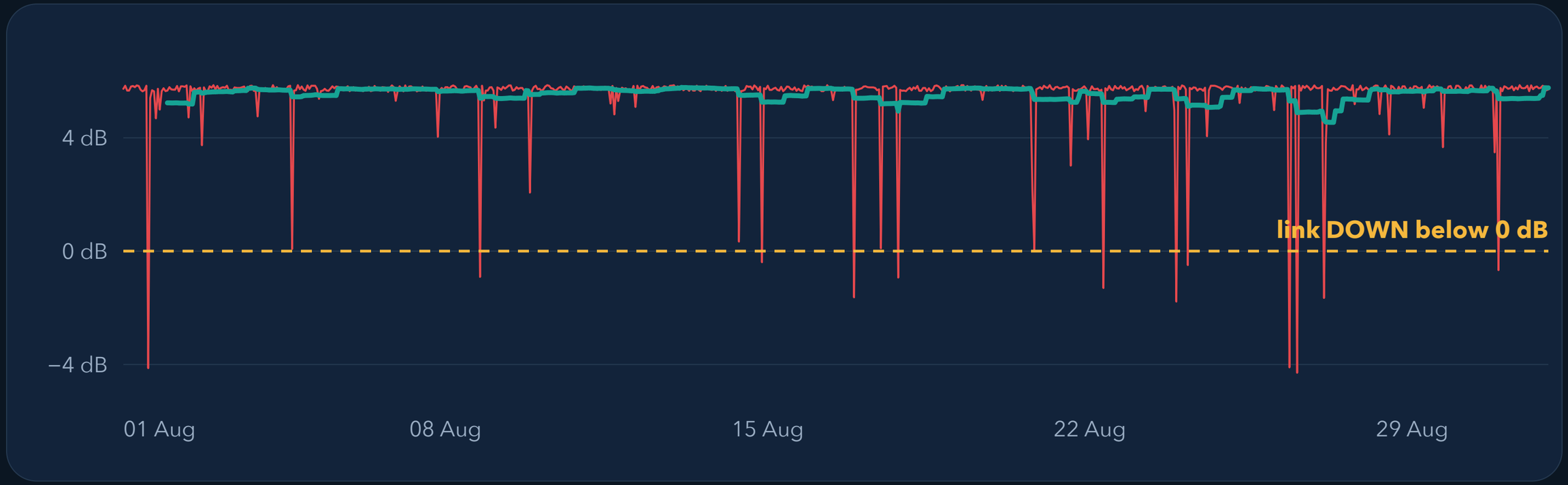
Low alpha (0.05)

waits for confirmation: very smooth, reacts slowly

```
s.ewm(alpha=0.5).mean()
```

THE MOST EXPENSIVE MISTAKE

Never report an SLA from a smoothed line 🚫



12 hours

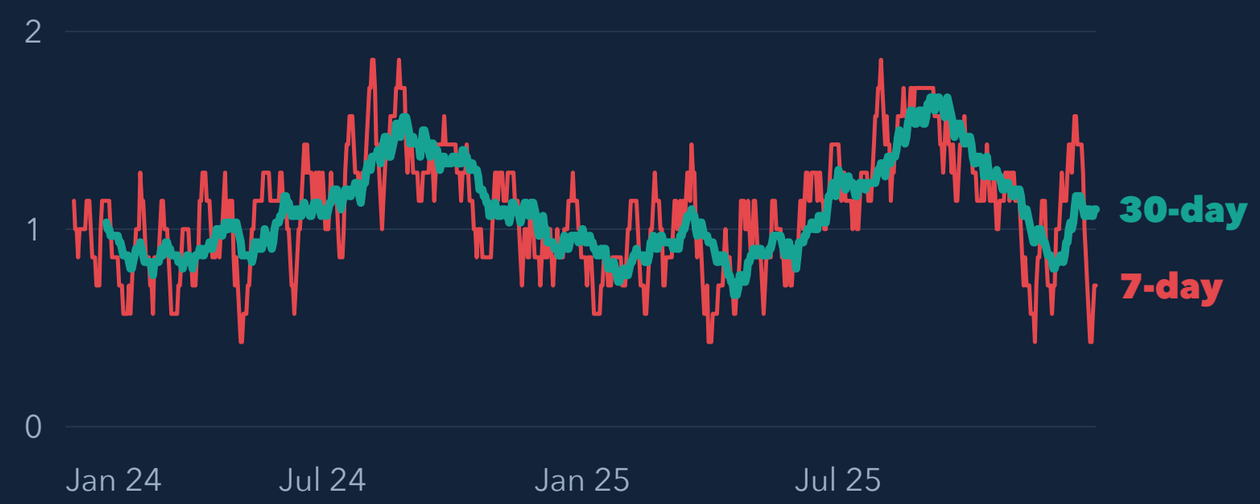
red hourly line: the link was really DOWN

0 hours

green 24-hour average: never below 4.5 dB

SMALL COUNTS NEED WIDER WINDOWS

Priority-1 tickets: rare and jumpy 🚨



Priority-1 tickets per day, rolling average

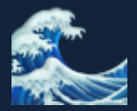
1.1
P1 tickets on an average day

17%
of days had zero P1 tickets

With 0, 1 or 2 tickets a day, a 7-day line still jumps. The 30-day line shows the real level.

TWO DIFFERENT QUESTIONS

`.rolling(7)` vs `.resample("W")`



rolling(7).mean()

"What's the average of the **last 7 days**, as of today?"

one value per DAY | 730 rows



resample("W").sum()

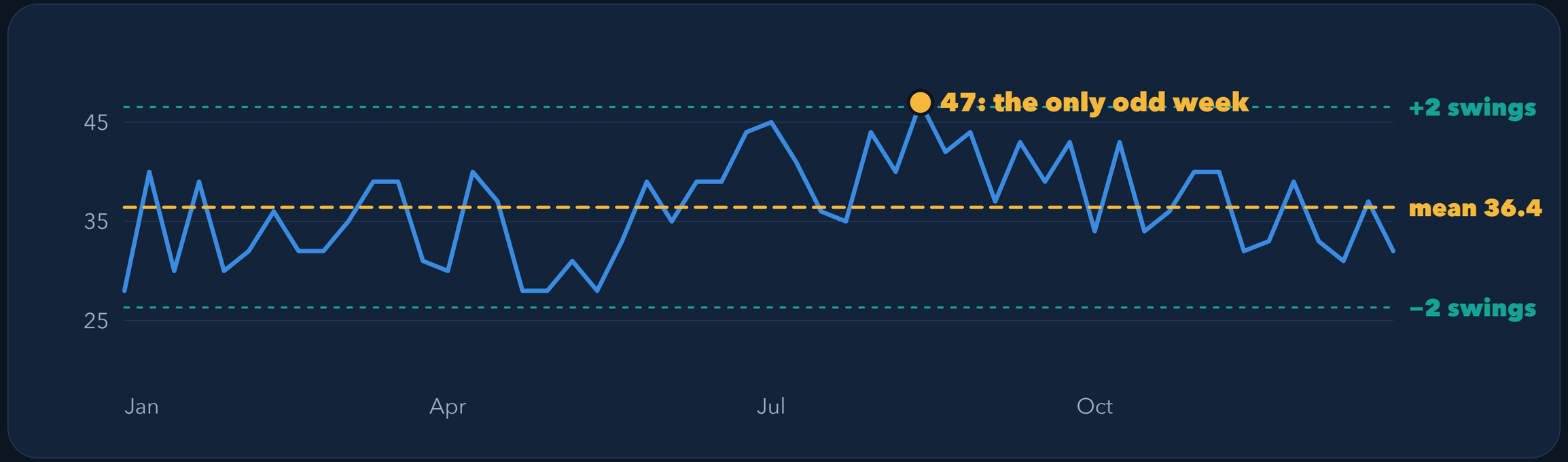
"How many tickets in **each calendar week**?" Days go into weekly buckets.

one value per WEEK | 105 rows

⚠️ Watch the edges: the last bucket holds only 2 days, so it shows 9 tickets. It's not a quiet week, just an incomplete one.

NOISE: DON'T CALL A MEETING FOR IT

38 vs 41 tickets = the same week twice



noc_tickets_weekly · 52 weeks

mean 36.4 tickets/week

typical swing ± 5

only 1 odd week of 52

YOUR TURN · 60 MINUTES

Open the exercise notebook 👩🏾‍💻

1. Click **Open in Colab** on W1D3-exercise
2. **File → Save a copy in Drive** 💾
3. Run **STEP 0**, then go top to bottom
4. Replace every ____, run, get the ✅



14 checks to pass

Part A: the weekly rhythm
Part B: windows and their price
🌶️ Bonus: spot the odd week

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

🌊 **Rolling window:** average the last N rows, slide forward. **7 days** cancels the weekly rhythm.

⏳ **Trailing** is safe but ~half a window late. **Centred** peeks at the future: reports only.

🚫 **Smoothing hides short events.** SLAs come from raw readings, never from an average.

Next week 👉 **Take It Apart**: splitting any series into trend, season and noise. 🧩